

Prompting

- A prompt is a text string that a user issues to a language model to get the model to do something useful.
- In prompting, the user's prompt string is passed to the language model, which iteratively generates tokens conditioned on the prompt.
- Thus the prompt creates a context that guides LLMs to generate useful outputs to achieve some user goal.

Example: sentiment classification task

- Sample Hotel Review

- Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relax.

- A prompt consisting of a review plus an incomplete statement

- Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relax. In short, our stay was

- Hotel Review Completions
- Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relaxed. In short our stay was
 - ... not a pleasant one. The staff at the front desk were not welcoming or friendly, and seemed disinterested in providing good customer service.
 - ... uncomfortable and not worth the price we paid. We will not be returning to this hotel.

- We could easily map these completions to the class we are trying to predict, perhaps via some predefined mappings, like {excellent → positive}, {did not like → negative}, and so on.

Power of doing many downstream tasks

- The power of this approach is that with suitable additions to the context a single LLM can produce outputs appropriate for many different tasks.
- For example, given a review we might want any of the following:
 - A summary,
 - Whether the review was truthful or likely to have been fabricated,
 - A translation to another language.

Template

- If we want to solve general tasks like summarization or translation, we don't want to have to create a new prompt each time we do the task.
- Instead the first step in prompting is to design one or more templates: task-specific prompting text along with slots for the particular input that is being processed.

Basic Prompt Templates

Summarization {input} ; tldr;

Translation {input} ; translate to French:

Sentiment {input}; Overall, it was

Fine-Grained-Sentiment {input}; What aspects were important in this review?

- An even more constraining style of prompt can specify the set of possible answers in the prompt. For example here is a prompt template to do sentiment analysis that prespecifies the potential answers:

A prompt consisting of a review plus an incomplete statement

Human: Do you think that “input” has negative or positive sentiment?

Choices:

(P) Positive

(N) Negative

Assistant: I believe the best answer is: (

LLM Outputs for Basic Prompts

Original Review (\$INPUT)	Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relax and away from the city life.
Sentiment	Prompt: \$INPUT + In short, our stay was Output: not enjoyable
Fine-grained Sentiment	Prompt: \$INPUT + These aspects were important to the reviewer: Output: 1. Poor service 2. Unpleasant location 3. Noisy and busy area
Summarization	Prompt: \$INPUT + tl;dr Output: I had a bad experience with the hotel's service and the location was loud and busy.
Translation	Prompt: \$INPUT + Translate this to French Output: Je n'ai pas aimé le service qui m'a été offert lorsque je suis entré dans l'hôtel. Je n'ai également pas aimé la zone dans laquelle se trouvait l'hôtel. Trop de bruit et d'événements pour que je me sente détendu et loin de la vie citadine.

Learning from Demonstrations: Few-Shot Prompting

- It's often possible to improve a prompt by including some labeled examples in the prompt template.
- We call such examples demonstrations.
- The task of prompting with examples is sometimes called few-shot prompting, as contrasted with zero-shot prompting which means instructions that don't include labeled examples.

Definition: This task is about writing a correct answer for the reading comprehension task. Based on the information provided in a given passage, you should identify the shortest continuous text span from the passage that serves as an answer to the given question. Avoid answers that are incorrect or provides incomplete justification for the question.

Passage: Beyoncé Giselle Knowles-Carter (born September 4, 1981) is an American singer, songwriter, record producer and actress. Born and raised in Houston, Texas, she performed in various singing and dancing competitions as a child, and rose to fame in the late 1990s as lead singer of R&B girl-group Destiny's Child. Managed by her father, Mathew Knowles, the group became one of the world's best-selling girl groups of all time. Their hiatus saw the release of Beyoncé's debut album, *Dangerously in Love* (2003), which established her as a solo artist worldwide, earned five Grammy Awards and featured the Billboard Hot 100 number-one singles "Crazy in Love" and "Baby Boy".

Examples:

Q: In what city and state did Beyoncé grow up?

A: Houston, Texas

Q: What areas did Beyoncé compete in when she was growing up?

A: singing and dancing

Q: When did Beyoncé release *Dangerously in Love*?

A: 2003

Q: When did Beyoncé start becoming popular?

A: